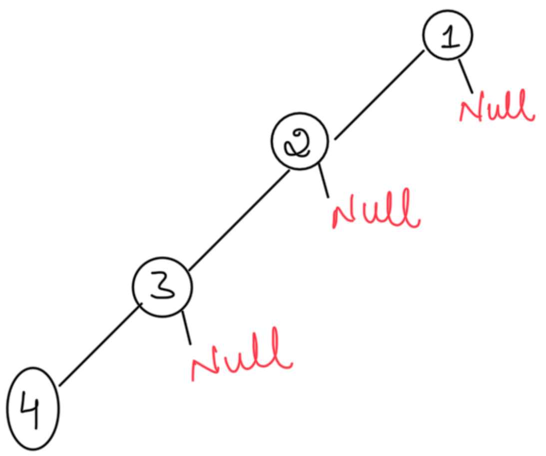
TREES - 2

DAY 164
21/Dec/2025

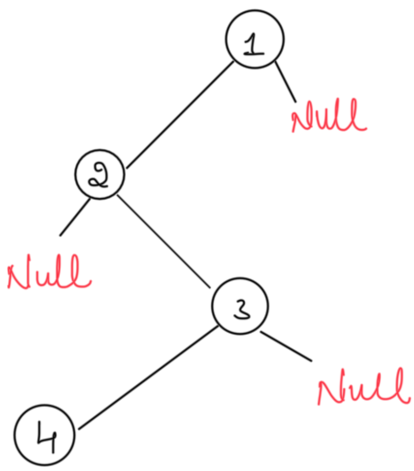
Skew trees

If every node in a tree has only one child (except leaf nodes) then we call such trees skew trees. If every has only left child then we call them left skew trees.

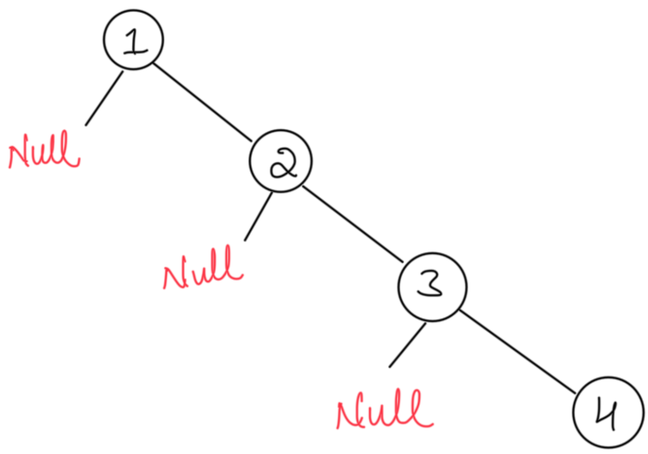
Similarly, if every node has only right child then we call them right skew trees.



Left skew Tree



Skew Tree

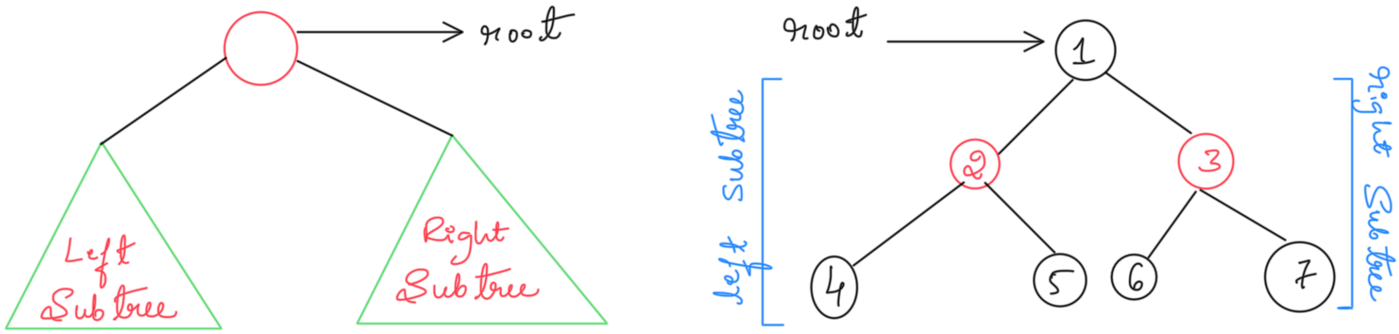


Right Skew Tree

Binary Trees

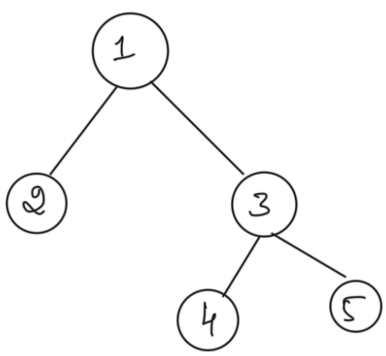
A tree is called Binary tree if Each node has Zero child, one child or two children. Empty tree is also a Valid Binary tree. We can Visualize a Binary tree as Consisting of a root and two disjoint Binary trees, called the left and right Subtrees of the root

Generic Binary Tree

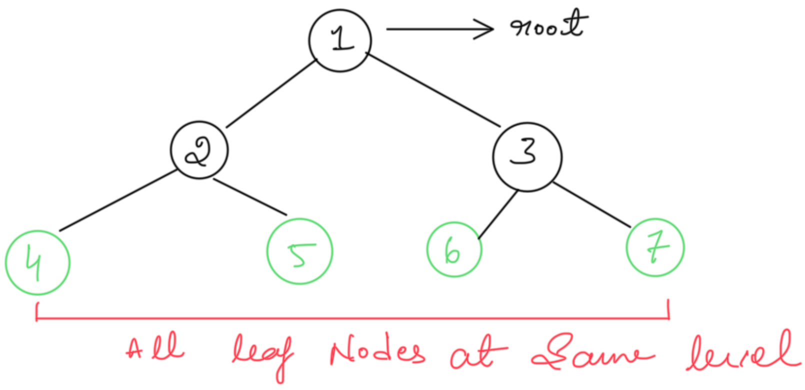


Types of Binary Trees

Strict Binary Tree : A Binary tree is called strict Binary tree if Each node has Exactly two children or no Children.

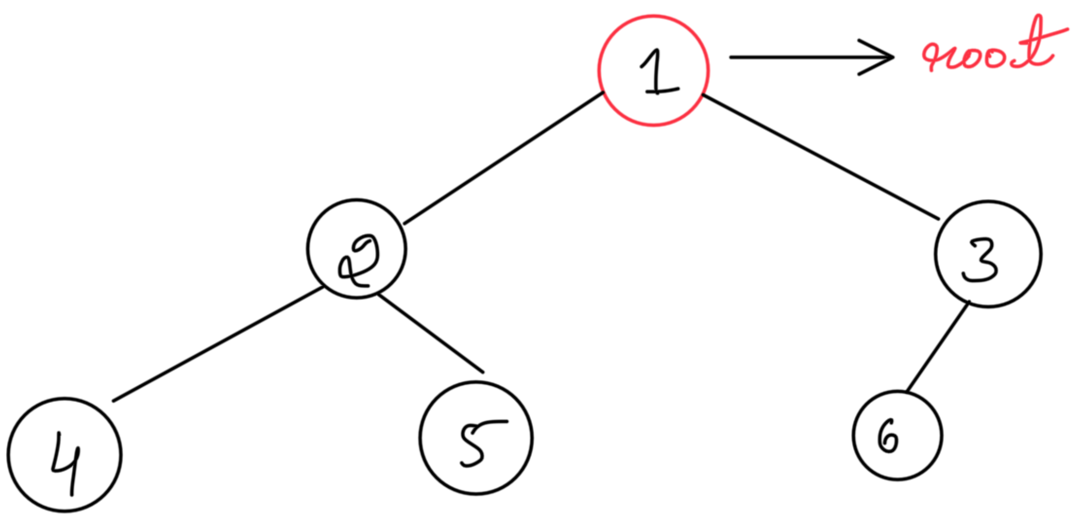


Full Binary Tree : A Binary tree is called as a full Binary tree if Each node has Exactly two Children and all leaf nodes are at the Same level.



Complete Binary Tree : Before defining the complete Binary tree let us assume that the height of the Binary tree is h.

In Complete Binary trees, if we give numbering for the nodes By starting at the root (let us say the root node has 1) then we get a complete Sequence from 1 to the number of nodes in the tree, while traversing we should give Numbering for Null pointers also. A Binary tree is called Complete Binary tree if all leaf nodes are at height 'h' or 'h-1' and also without any missing numbers in Sequence



1️⃣ Is a Singly Linked List a Tree?

❌ **No**

Why?

- A **tree** is hierarchical (parent → children)
- A **singly linked list** is linear (one node → next node)

Singly Linked List

- Each node has **only one pointer** → `next`
- No concept of left/right child
- No branching

```text
10 → 20 → 30 → 40 → NULL
```

So structurally and conceptually → **NOT a tree**

3️⃣ So... Is a Skewed Tree a Linked List?

Conceptually ❌ NO

Structurally ⚠️ ALMOST YES

Aspect	Linked List	Skewed Tree
Node pointers	`next`	`left` / `right`
Hierarchy	❌ No	✅ Yes
Traversal	Linear	Tree traversal
Tree rules	❌	✅

👉 **Important exam/interview line:**

> A skewed binary tree is *structurally similar* to a linked list but **logically it is still a tree.**

✔ **A singly linked list is not a tree.**

✔ **A left or right skewed tree resembles a linked list but remains a tree due to its hierarchical structu**